

JEE MAIN REVISION MASTER: INDEFINITE INTEGRALS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 18 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Indefinite Integrals - Core Application	18 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Indefinite Integrals - Core Application (Very High Yield — 18 PYQs)

Core Principle: Key problem pattern in Indefinite Integrals.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (Numerical): "If $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = -\frac{p_1 q_1}{c_1} (\cot x)^{p_2 q_2} (\csc x)^{p_3 q_3} - \frac{p_4 q_4}{c_4} (\cot x)^{p_5 q_5} - \frac{p_6 q_6}{c_6} (\cot x)^{p_7 q_7} - \frac{p_8 q_8}{c_8} (\cot x)^{p_9 q_9} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4$ is equal to ____" [Ans: 16]
- [2025 Evening Shift] Q#2 (Numerical): "If $\int (\frac{1}{x} + \frac{1}{x^3}) (\sqrt{23}) dx = -\frac{\alpha}{3(\alpha+1)} (3x^\alpha \ln \beta + x^\alpha \gamma)^{\frac{\alpha+1}{\alpha}} + C, x > 0$, then $\alpha + \beta + \gamma$ is ____" [Ans: 19]
- [2025 Evening Shift] Q#3 (Numerical): "If $\int \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2} - x} dx = \frac{1}{m} (\sqrt{1+x^2} + x)^n (\sqrt{1+x^2} - x) + C$ where C is the constant of integration, then $m \cdot n$ is ____" [Ans: 379]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Morning Shift (Q#1)

If $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = -\frac{p_1 q_1}{c_1} (\cot x)^{p_2 q_2} - \frac{p_3 q_3}{c_3} (\cot x)^{p_4 q_4} - \frac{p_5 q_5}{c_5} (\cot x)^{p_6 q_6} - \frac{p_7 q_7}{c_7} (\cot x)^{p_8 q_8} + C$, where p_i and q_i are positive integers with $\gcd(p_i, q_i) = 1$ for $i = 1, 2, 3, 4$ and C is the constant of integration, then $15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4$ is equal to ____

Correct Answer: 16

Solution Analysis: $\int (\sin x)^{-11/2} (\cos x)^{-5/2} dx = \int \frac{1}{\sin^{11/2} x \cos^{5/2} x} dx = \int \frac{1}{\sin^9 x \cos^5 x} dx$
Put $\tan x = t$, then $dx = \frac{dt}{1+t^2}$
 $\int \frac{1}{\sin^9 x \cos^5 x} \cdot \frac{dt}{1+t^2} = \int \frac{1}{(1+t^2)^{5/2} (1+t^2)^{3/2}} dt = \int \frac{1}{(1+t^2)^4} dt$
Let $1+t^2 = u$, then $2t dt = du$
 $\int \frac{1}{u^4} \cdot \frac{du}{2} = -\frac{1}{2 \cdot 3} u^{-3} + C = -\frac{1}{6} (1+t^2)^{-3} + C = -\frac{1}{6} (\sec^2 x)^{-3} + C = -\frac{1}{6} \cos^6 x + C$
Comparing with the given form, we get $p_1 = 1, q_1 = 1, p_2 = 3, q_2 = 2, p_3 = 1, q_3 = 1, p_4 = 1, q_4 = 1$
Hence, $15p_1 p_2 p_3 p_4 q_1 q_2 q_3 q_4 = 15 \cdot 1 \cdot 3 \cdot 1 \cdot 1 \cdot 1 \cdot 2 \cdot 1 \cdot 1 = 16$